

Two-Dimensional Titanium Carbide MXene As a Cathode Material for Hybrid Magnesium/Lithium-Ion Batteries

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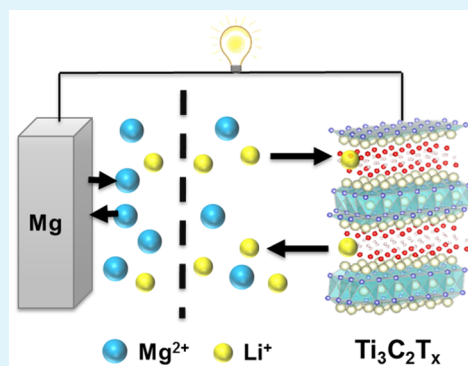
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Supporting Information

ABSTRACT: As an alternative to pure lithium-ion, Li^+ systems, a hybrid magnesium, Mg^{2+} , and Li^+ battery can potentially combine the high capacity, high voltage, and fast Li^+ intercalation of Li-ion battery cathodes and the high capacity, low cost, and dendrite-free Mg metal anodes. Herein, we report on the use of two-dimensional titanium carbide, $\text{Ti}_3\text{C}_2\text{T}_x$ (MXene), as a cathode in hybrid $\text{Mg}^{2+}/\text{Li}^+$ batteries, coupled with a Mg metal anode. Free-standing and flexible $\text{Ti}_3\text{C}_2\text{T}_x$ /carbon nanotube composite “paper” delivered $\sim 100 \text{ mAh g}^{-1}$ at 0.1 C and $\sim 50 \text{ mAh g}^{-1}$ at 10 C. At 1 C the capacity was maintained for >500 cycles at 80 mAh g^{-1} . The Mo_2CT_x MXene also demonstrated good performance as a cathode material in this hybrid battery. Considering the variety of available MXenes, this work opens the door for exploring a new large family of 2D materials with high electrical conductivity and large intercalation capacity as cathodes for hybrid $\text{Mg}^{2+}/\text{Li}^+$ batteries.

KEYWORDS: 2D material, titanium carbide, MXene, hybrid $\text{Mg}^{2+}/\text{Li}^+$ battery, energy storage



INTRODUCTION

In recent years, the development of rechargeable batteries beyond Li-ion based systems has been attracting much attention because of the increasing energy storage demands in terms of energy density, safety, and cost.¹ In particular, tremendous efforts have been devoted to batteries that make use of multivalent ions, such as magnesium (Mg^{2+}) and aluminum (Al^{3+}) ions. The latter can lead to higher energy densities, and are cheaper and more abundant than Li salts.^{2,3} In general, the batteries would also be safer. For example, Mg is safe to handle in ambient atmospheres, abundant in nature and cheaper than Li metal. Because of its dendrite-free deposition/dissolution characteristics, Mg can directly serve as an anode in Mg-ion battery systems with a much higher theoretical capacity (gravimetric, 2205 mAh g^{-1} ; volumetric, 3832 mAh cm^{-3}) compared with other typical anode materials.² However, the kinetically sluggish Mg intercalation/insertion and diffusion results in very few choices of cathode materials. This poor rate capability, in turn, has hindered the practical application of Mg-ion batteries.^{4–7}

As an alternative to developing cathode materials for the intercalation of Mg ions, the concept of hybrid $\text{Mg}^{2+}/\text{Li}^+$ battery system, comprised of a Mg anode, a Li^+ ion intercalation cathode, and a dual-salt electrolyte, with both

Mg^{2+} and Li^+ ions, has been proposed.^{8–10} A hybrid $\text{Mg}^{2+}/\text{Li}^+$ battery combines the high capacity, high voltage, and fast Li^+ intercalation of Li-ion cathodes and the high capacity, low cost, and dendrite-free Mg anodes. The Chevrel phase, Mo_6S_8 , was the first reported and widely investigated cathode material for hybrid $\text{Mg}^{2+}/\text{Li}^+$ batteries.^{11,12} Significant improvements in capacity and rate capabilities were demonstrated when compared with Mg-ion battery systems in which no Li^+ -containing salts were added to the electrolyte.⁹ Following that, lithium iron phosphate (LiFePO_4), titanium disulfide (TiS_2), and titanium dioxide (TiO_2) were also employed as cathode materials in hybrid $\text{Mg}^{2+}/\text{Li}^+$ battery systems.^{5,10,13,14} Note that the simple use of a conventional Li-ion battery cathode does not necessarily guarantee a reversible electrochemical reaction coupled with superior performance, since the compatibility between the cathode materials and the complex known electrolytes for hybrid $\text{Mg}^{2+}/\text{Li}^+$ batteries have to be taken

Special Issue: New Materials and Approaches for Beyond Li-ion Batteries

Received: April 8, 2016

Accepted: May 31, 2016

Published: June 8, 2016

into account.¹⁰ As a result, the choice of cathode materials for high-performance hybrid $\text{Mg}^{2+}/\text{Li}^{+}$ batteries is still limited.

Herein we report, for the first time, on how two-dimensional (2D) titanium carbide, $\text{Ti}_3\text{C}_2\text{T}_x$ (MXene), performs as a cathode in a hybrid $\text{Mg}^{2+}/\text{Li}^{+}$ battery. MXenes are a family of 2D transition metal carbides/carbonitrides.^{15,16} They have the formula $\text{M}_{n+1}\text{X}_n\text{T}_x$, where M is an early transition metal, such as Ti, Nb, V, Ta, Cr, and Mo, X is carbon and/or nitrogen, $n = 1, 2$ or 3 , and T_x refers to surface functional groups such as OH, O, and F. MXenes offer metallic conductivity, hydrophilic surfaces, coupled with excellent mechanical properties. They also show superior capacity for reversible intercalation of many organic molecules and metal cations (e.g., Li^{+} , Na^{+} , Mg^{2+} , Al^{3+} , etc.).^{17–19} So far, MXenes have been tested in a variety of energy storage devices, including supercapacitors, Li- and Na-ion batteries, and Li–S batteries.^{16,19–24}

RESULTS AND DISCUSSION

Figure 1 shows a schematic of a hybrid $\text{Mg}^{2+}/\text{Li}^{+}$ battery in which $\text{Ti}_3\text{C}_2\text{T}_x$ is the cathode, Mg is the anode, and an all-

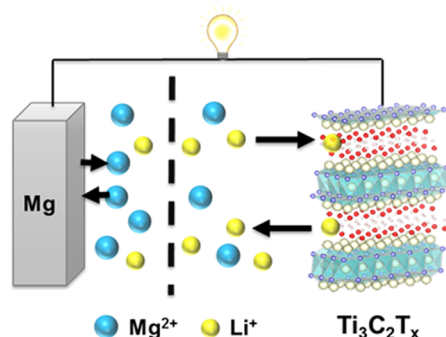


Figure 1. Schematic of a hybrid $\text{Mg}^{2+}/\text{Li}^{+}$ battery using $\text{Ti}_3\text{C}_2\text{T}_x$ MXene as the cathode material.

phenyl complex (APC)-LiCl in tetrahydrofuran (THF) is the electrolyte. $\text{Ti}_3\text{C}_2\text{T}_x$ was chosen because it is the best studied MXene to date. At $> 2000 \text{ S/cm}$, its electrical conductivity is quite high.^{16,20} Note that reversible Mg deposition/dissolution would occur at the anode before Li because of its higher thermodynamic redox potential. Meanwhile at the cathode, Li^{+} ions would dominate the intercalation process because Mg^{2+} diffusion is several orders of magnitude slower than Li^{+} .¹⁴

Both multilayered and delaminated $\text{Ti}_3\text{C}_2\text{T}_x$ were evaluated as cathode materials. Details of materials synthesis can be found in the Supporting Information. The as-synthesized multilayer $\text{Ti}_3\text{C}_2\text{T}_x$ (ML- $\text{Ti}_3\text{C}_2\text{T}_x$) powder is composed of stacked $\text{Ti}_3\text{C}_2\text{T}_x$ flakes, with a morphology (Figure 2a) reminiscent of exfoliated graphite. After delamination, a colloidal solution of delaminated $\text{Ti}_3\text{C}_2\text{T}_x$ flakes was obtained, in which single or few-layer $\text{Ti}_3\text{C}_2\text{T}_x$ flakes one to a few nanometers thick with lateral dimensions ranging from hundreds of nanometers to several micrometers exist (Figure 2b). This delaminated MXene, henceforth referred to as d- $\text{Ti}_3\text{C}_2\text{T}_x$, can in turn be fabricated into free-standing and flexible “paper” with the addition of carbon nanotubes (CNTs) by a simple vacuum-assisted filtration method (Figure S1).²¹ The CNTs serve as spacers between the d- $\text{Ti}_3\text{C}_2\text{T}_x$ flakes and prevent their compact restacking. If the flakes restack, rapid diffusion of electrolyte ions between the MXene flakes can be severely hindered.^{21,25}

A cross-sectional scanning electron microscope (SEM) image of the as-fabricated d- $\text{Ti}_3\text{C}_2\text{T}_x/\text{CNT}$ composite paper is shown in Figure 2c, from which it is obvious that the film’s thickness is $\sim 5 \mu\text{m}$ and that the structure is more open when compared to pure $\text{Ti}_3\text{C}_2\text{T}_x$.²¹ Figure 2d shows the X-ray diffraction (XRD) patterns of the ML- $\text{Ti}_3\text{C}_2\text{T}_x$ and d- $\text{Ti}_3\text{C}_2\text{T}_x/\text{CNT}$ paper. The (002) peak of the ML- $\text{Ti}_3\text{C}_2\text{T}_x$ shifts from 7.3 to 6.2° for the d- $\text{Ti}_3\text{C}_2\text{T}_x/\text{CNT}$ paper, indicating an increase in c -lattice parameter (c -LP) from 24.2 to $28.4 \text{ \AA}$.

The ML- $\text{Ti}_3\text{C}_2\text{T}_x$ -based cathode was prepared by mixing 80 wt % ML- $\text{Ti}_3\text{C}_2\text{T}_x$ particles, 10 wt % carbon black and 10 wt % polytetrafluoroethylene (PTFE) binder. The d- $\text{Ti}_3\text{C}_2\text{T}_x/\text{CNT}$ paper was directly used as the cathode. Both cathodes were assembled in coin cells without the use of current collectors. The cyclic voltammetric (CV) curves in Figure 3a show broad Li^{+} ion intercalation/extraction peaks for both ML- $\text{Ti}_3\text{C}_2\text{T}_x$ and d- $\text{Ti}_3\text{C}_2\text{T}_x/\text{CNT}$ electrodes. Like in our previous work,¹⁸ the latter showed a much improved capacity compared to the former. At a current density of 1 C , the ML- $\text{Ti}_3\text{C}_2\text{T}_x$ cathode showed stable discharge capacities around 18 mAh g^{-1} (Figure 3b). Although a low discharge capacity of $\sim 5 \text{ mAh g}^{-1}$ was initially obtained, when the d- $\text{Ti}_3\text{C}_2\text{T}_x/\text{CNT}$ paper electrode was cycled, at 1 C , it increased rapidly upon cycling (Figure S2). After ~ 50 cycles, the capacity was stable at $\sim 80 \text{ mAh g}^{-1}$ (Figure S2). This increase in capacity is attributable to improved electrolyte accessibility upon cycling, which is common for MXene-based paper electrodes, as well as for other film electrodes.^{26,27} Such an increase in capacity was not observed for the ML- $\text{Ti}_3\text{C}_2\text{T}_x$ electrodes, indicating that electrolyte ion accessibility in this case was not improved during cycling.

The cycling profile of a d- $\text{Ti}_3\text{C}_2\text{T}_x/\text{CNT}$ paper electrode, precycled 100 times, is shown in Figure 3b. Discharge capacities of $\sim 80 \text{ mAh g}^{-1}$ were retained for up to 500 cycles, with Coulombic efficiencies close to 100% (see right y-axis in Figure 3b). Compared to the ML- $\text{Ti}_3\text{C}_2\text{T}_x$ electrodes, the paper ones, exhibited significant higher capacities, presumably because the $\text{Ti}_3\text{C}_2\text{T}_x$ flakes after delamination were more accessible to the Li ions.^{18,25} Figure 3c shows the rate performance of both ML- $\text{Ti}_3\text{C}_2\text{T}_x$ and d- $\text{Ti}_3\text{C}_2\text{T}_x/\text{CNT}$ electrodes. At 0.1 C , the ML- $\text{Ti}_3\text{C}_2\text{T}_x$ cathode delivered discharge capacities around 40 mAh g^{-1} ; no capacity was observed at current rates higher than 5 C . However, the d- $\text{Ti}_3\text{C}_2\text{T}_x/\text{CNT}$ electrodes delivered discharge capacities of $\sim 105 \text{ mAh g}^{-1}$ at 0.1 C . At 5 and 10 C , capacities of ~ 65 and $\sim 40 \text{ mAh g}^{-1}$ were recorded, respectively. These results show that the rate performance is significantly improved after delamination and if the flakes are prevented from restacking with the CNTs. Note that the contribution of CNTs to the overall capacities is negligible because of their limited specific capacity and low content in the composite “paper”.²⁵

At 0.1 C , there was no need for precycling for the d- $\text{Ti}_3\text{C}_2\text{T}_x/\text{CNT}$ electrodes to achieve good performance. This is probably because there was enough time for Li^{+} diffusion into the electrodes during each cycle at such a low current densities. Furthermore, and in sharp contrast to our previously reported $\text{Ti}_3\text{C}_2\text{T}_x$ work,^{18,27} no obvious first-cycle irreversible capacities was observed for either the ML- $\text{Ti}_3\text{C}_2\text{T}_x$ or d- $\text{Ti}_3\text{C}_2\text{T}_x/\text{CNT}$ cathodes at 0.1 C (Figure 3c). The main reason for this state of affairs is that the formation of a solid electrolyte interphase (SEI) was prevented in the voltage window tested, viz. $> 0.2 \text{ V}$ vs Mg/Mg^{2+} or $> 0.9 \text{ V}$ vs Li/Li^{+} .^{5,9,10} This is clearly one advantage of using this hybrid $\text{Mg}^{2+}/\text{Li}^{+}$ battery configuration.

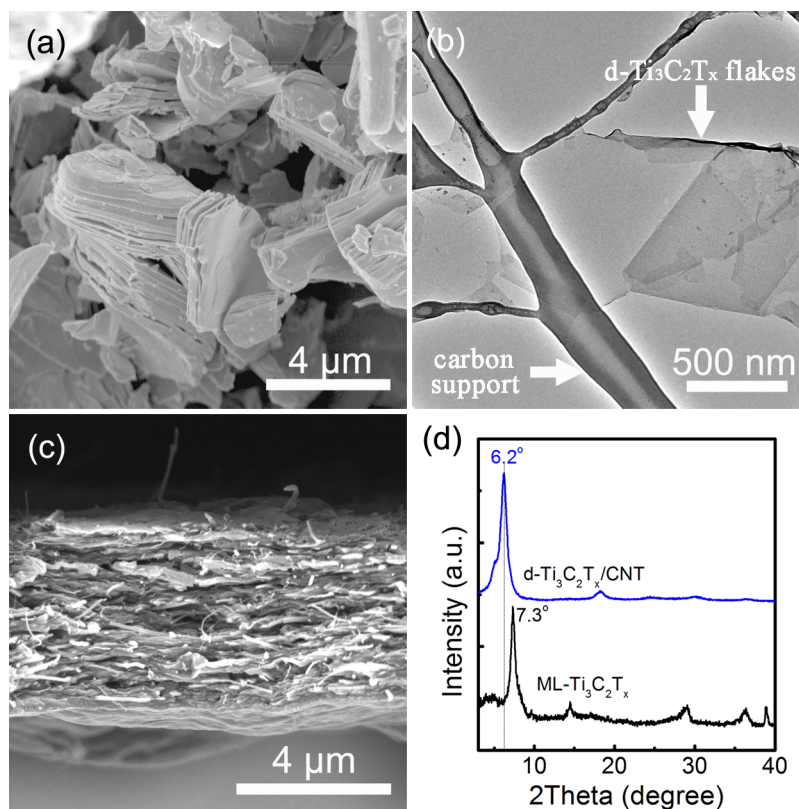


Figure 2. (a) SEM image of ML-Ti₃C₂T_x particles; (b) TEM image of d-Ti₃C₂T_x flakes; (c) cross-sectional SEM image of a d-Ti₃C₂T_x/CNT composite film; (d) XRD patterns of ML-Ti₃C₂T_x and d-Ti₃C₂T_x/CNT composite paper.

The charge–discharge profiles of the d-Ti₃C₂T_x/CNT cathode at different current rates are shown in Figure 3d. The discharge capacities initiated from potentials <1.7 V vs Mg/Mg²⁺. No obvious plateaus were observed in any of the charge–discharge curves which can be attributed to the 2D nature of our d-Ti₃C₂T_x flakes that in turn allow fast diffusion of the Li⁺ ions.²⁸

It is important to note here that without LiCl addition to the APC electrolyte, the capacity of the d-Ti₃C₂T_x/CNT cathode was quite negligible (Figure S3), proving that it is the Li⁺, and not the Mg²⁺, cations that are responsible for the measured capacities (Figure S3). Post analysis of discharged d-Ti₃C₂T_x/CNT electrodes (after 100 cycles) was carried out to provide a better understanding on the charge–discharge process. Transmission electron microscope (TEM) micrographs (Figure S4b) revealed that the morphologies of d-Ti₃C₂T_x flakes were well preserved after cycling. The composition of the tested d-Ti₃C₂T_x flakes were analyzed by energy-dispersive X-ray spectroscopy (EDX) and showed no appreciable Mg content (Figure S4c). Therefore, it is evident that only Li⁺ ion intercalated the cathode in this hybrid cell, which agrees well with previous work on Mo₆S₈ and TiS₂ cathodes.^{9,10} XRD patterns showed a significant increase of 11.7 Å in the *d*-spacing (where *d* = *c*-LP/2) of the Ti₃C₂T_x flakes after cycling (Figure S4a). This is common for MXenes and can account at least in part for the capacity increase during cycling mentioned above.¹⁹

The capacities obtained herein for the d-Ti₃C₂T_x/CNT cathodes are comparable to the Chevrel phase and TiS₂ cathodes, more commonly used for Mg²⁺/Li⁺ hybrid batteries.^{9,10} However, since modeling and experimental results suggest that other MXene compositions, such as Nb₂CT_x, V₂CT_x and Mo₂CT_x, should exhibit higher Li⁺ ion

capacities,^{16,29,30} we also tested Mo₂CT_x MXene in the same hybrid cells. At around ~120 mAh g^{−1}, higher capacities were achieved at 0.1 C, although the rate capability was poorer presumably due to the relatively lower conductivities of the Mo₂CT_x (Figure S5). The same should be true for Li- and Na-ion batteries.^{16,29} This indicates that the performance of MXene-based electrodes can be improved by optimizing the MXene compositions and/or electrode architecture.^{27,30} The mixing of high-capacity (e.g., Mo₂CT_x) and high conductivity (e.g., Ti₃C₂T_x) MXenes may also be a promising route to explore higher performance electrodes.

CONCLUSIONS

In summary, we demonstrated that Ti₃C₂T_x can be used as a cathode in hybrid Mg²⁺/Li⁺ batteries. A capacity of ~105 mAh g^{−1} at 0.1 C was obtained with free-standing and flexible d-Ti₃C₂T_x/CNT paper cathodes. Remarkable capacity retention at high current rates and long cycling were demonstrated.

MXenes with other compositions, such as Mo₂CT_x, were also compatible with hybrid Mg²⁺/Li⁺ battery electrolytes and exhibited higher capacities as cathode materials. Considering the variety of MXenes,¹⁶ this work suggests that a large family of cathode materials with high performance are available for the hybrid Mg²⁺/Li⁺ batteries.

Performance comparisons of MXenes, with diverse compositions under different cationic environments, such as Li⁺, Na⁺, and Mg²⁺ batteries, would be helpful to understand the differences of their intrinsic electrochemical properties and the intercalation mechanism in each system. This can greatly promote the development of hybrid Mg²⁺/Li⁺ battery systems and, thereby, benefit the field of electrochemical energy storage.

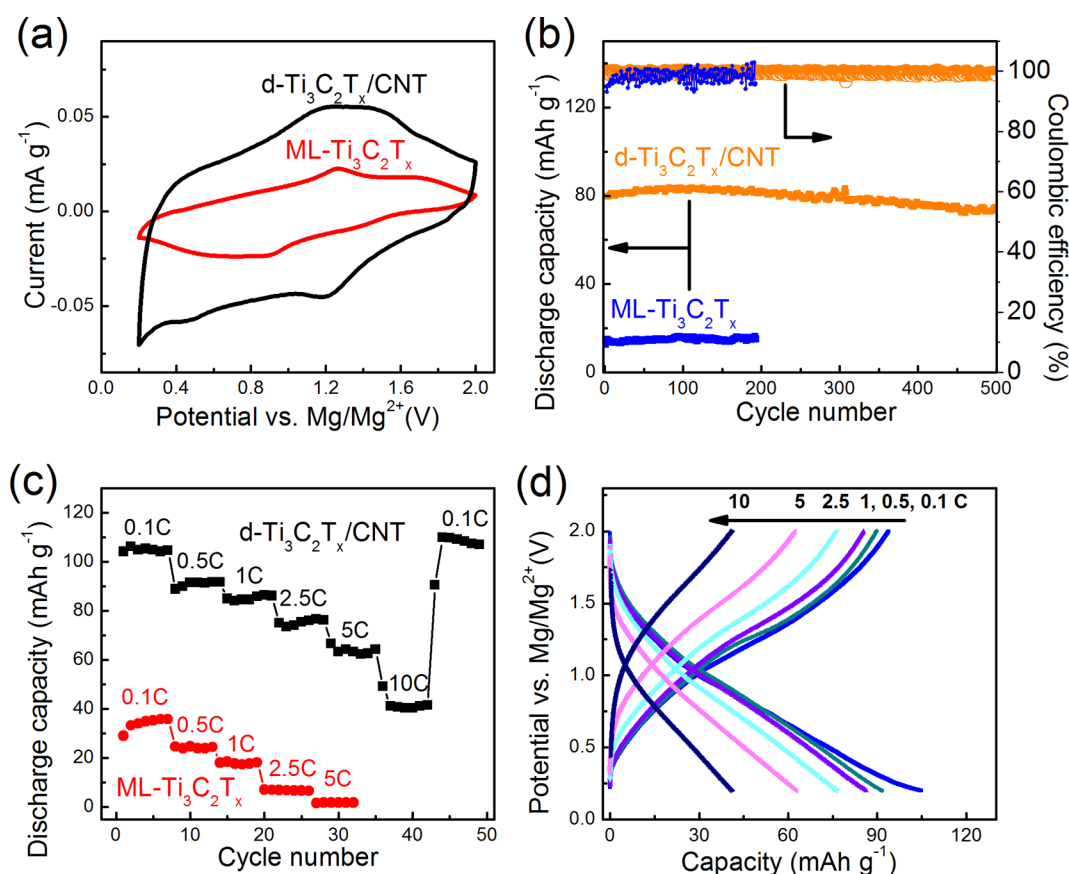


Figure 3. (a) Fifth CV cycles for ML-Ti₃C₂T_x and d-Ti₃C₂T_x/CNT cathodes tested at 0.2 mV s⁻¹; (b) cycle stability of ML-Ti₃C₂T_x and precycled -100 times - d-Ti₃C₂T_x/CNT cathodes at 1 C (1 C = 100 mA g⁻¹); (c) rate capability of ML-Ti₃C₂T_x and d-Ti₃C₂T_x/CNT cathodes; (d) charge/discharge profiles of a precycled d-Ti₃C₂T_x/CNT cathode at different current rates.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acsami.6b04198.

Detailed experimental methods on the synthesis, characterizations, and electrochemical measurements of the ML-Ti₃C₂T_x and d-Ti₃C₂T_x/CNT composites papers, as well as supplementary experimental results of the samples (PDF)

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Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

This work was supported by the Toyota Research Institute of North America and the Korean National Research Foundation via the NNFC-KAIST-Drexel Nano² Co-op Center (NRF-2015K1A4A3047100). A.B. acknowledges support from the Korea CCS R&D program funded by the Ministry of Science, ICT, and Future Planning (NRF-2014M1A8A1049297). P.U.

and S.K. were supported by the US National Science Foundation under grant number DMR-1310245. C.E.R. was supported by the Chinese Scholarship Council (CSC). The authors thank the Core Facilities of Drexel University for providing access to XRD, SEM, and TEM.

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